MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: I)
(October, 2025) OFFLINE mode

Subject Code: BAS 102	**Subject: Applied Physics**
Time : 1 ½ Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Find the value of the constant "p" so that the given vector is solenoidal. $A= (y^2-z^2+pyz-2x)\, i + (3xz+2xy)\, j + (3xy-2xz+2z)\, k$	
	b) In an interference pattern formed by two coherent sources, the maximum and minimum intensities are found to be 9 I_0 and I_0 respectively. What will be the individual wave intensities of the wave?	
	c) In free space $\bar{E} = 10 \cos(\omega t - 50x)\, i$ Find the value of Displacement current density (J_0) and Magnetic field vector (H).	
	d) At what angle will 650 nm light produce a second order maximum when falling on a grating whose slits are 1.2 × 10−3 cm apart?	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) Discuss the phenomena of Fraunhoffer diffraction at single slit with a well-labeled intensity distribution graph. Also, show the intensities of successive maxima are nearly in the ratio. $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2}$	5
	b) (i) Derive the conditions for constructive and destructive interference due to reflection when monochromatic light is incident on a thin film of thickness t and refractive index n. (ii) A parallel beam of sodium light of wavelength 5890×10^{-8}cm is incident on a thin glass plate of refractive index 1.5, such that the angle of refraction into the plate is 60∘. Calculate the smallest thickness of the plate which will make it appear dark by reflection	3.5, 1.5
	c) Explain the construction and working principle of a Nicol prism, and discuss its applications.	5
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Using Maxwell's equations, establish the wave equations for the transverse electric and magnetic fields in a non-conducting dielectric medium. Show that the speed of an electromagnetic wave in the dielectric medium is less than the speed of light in vacuum. Also, evaluate the Poynting vector.	5
	b) State and explain the modified form of Ampère's circuital law (Ampère-Maxwell law). Derive its expression and show how it accounts for the displacement current in a time-varying electric field. Discuss its significance in the propagation of electromagnetic waves.	5
	c) (i) Derive the continuity equation for electric charge and explain its physical significance. (ii) State and explain Poynting Theorem giving its physical significance.	2.5, 2.5